

Electronic Devices & Circuits Laboratory

Experiment 5: Closing the Loop — Unity-Gain and Gain-of-10 Feedback Configurations

1. Objective

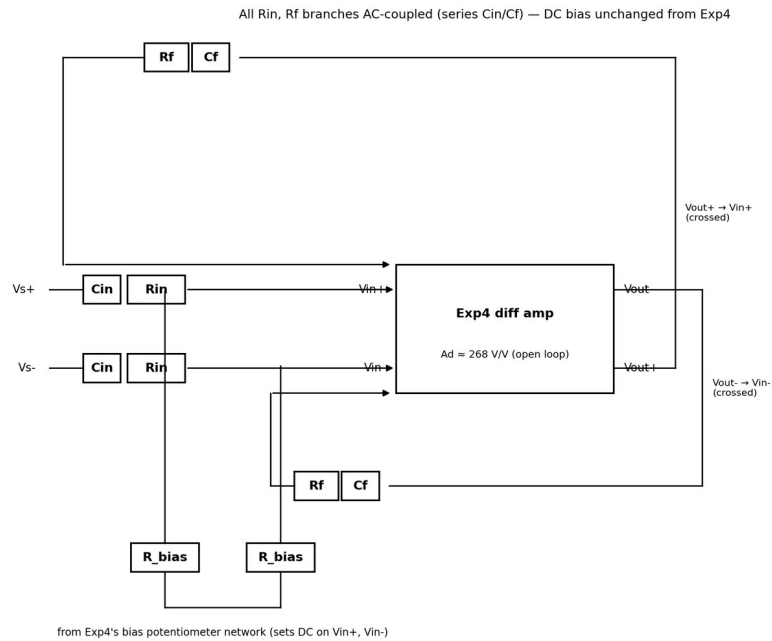
To apply negative feedback around the Experiment 4 differential amplifier (the same physical circuit, unchanged) using external resistor-capacitor networks, and to configure it first as a unity-gain stage and then as a gain-of-10 stage; to verify the closed-loop gain against the R_f/R_{in} prediction and against the finite-loop-gain-corrected prediction; and to observe the low-frequency roll-off set by the AC-coupling capacitors.

2. Components and Equipment Required

Item	Specification	Quantity
Experiment 4 circuit	SI9926BDY diff amp, fully assembled and biased ($V_{DD} = 12\text{ V}$, $R_D = 130\ \Omega$, $R_{SS} = 12\ \Omega$, $I_S \approx 80\text{ mA}$), unchanged	1
Feedback resistor R_f	$10\text{ k}\Omega$, $\pm 5\%$	2
Input resistor R_{in} (unity gain)	$10\text{ k}\Omega$, $\pm 5\%$	2
Input resistor R_{in} (gain of 10)	$1\text{ k}\Omega$, $\pm 5\%$	2
Coupling capacitor C_f	$1\ \mu\text{F}$ (non-polarized, or electrolytic with polarity noted — see Section 6)	2
Coupling capacitor C_{in} (unity gain)	$1\ \mu\text{F}$	2
Coupling capacitor C_{in} (gain of 10)	$10\ \mu\text{F}$	2
DC decoupling resistor R_{bias}	$1\text{ M}\Omega$ (isolates the AC network from the Exp4 bias pots)	2
Function generator	Sine output, 1 Hz – few kHz , low output impedance	1
Oscilloscope	2-channel, to view $V_{in,diff}$ and $V_{out,diff}$ amplitude and phase	1

3. Circuit Diagram

The feedback is crossed — V_{out+} feeds V_{in+} through one R_f/C_f branch and V_{out-} feeds V_{in-} through the other — because in this diff pair V_{out+} is non-inverting with respect to V_{in+} (increasing V_{in+} increases V_{out+} , since M1 steals current from M2's branch). Straight, uncrossed feedback would therefore be positive feedback; only the crossed connection shown here is stabilizing. All R_{in} and R_f branches are AC-coupled through series capacitors, and a large R_{bias} ($1\text{ M}\Omega$, negligible loading since it is $\gg R_{in}$, R_f) carries the DC bias from Experiment 4's potentiometer network to each gate, so the quiescent point set in Experiment 4 is undisturbed.



4. Theory

With the diff pair's open-loop differential gain A_d and a feedback network (R_f , R_{in}) applying negative feedback as shown, KCL at each (infinite-impedance) gate node gives the standard inverting-feedback result:

$V_{out,diff} / V_{s,diff} = -A_d \cdot R_f / (R_f + R_{in} \cdot (1 + A_d)) \approx -R_f / R_{in}$ for large loop gain $A_d \cdot \beta$, where $\beta = R_{in} / (R_f + R_{in})$

The closed-loop gain is inverting because of the crossed connection: $V_{out,diff} = -(R_f / R_{in}) \cdot V_{s,diff}$ in the ideal (high loop-gain) limit. The fraction by which the actual gain falls short of $-R_f / R_{in}$ is set by the loop gain $A_d \cdot \beta$ — the smaller the loop gain, the larger the gap, which is itself something to measure and explain, not an error to eliminate.

Each coupling capacitor forms a high-pass network with its series resistor, with corner frequency $f_c = 1/(2\pi RC)$. All four RC combinations here are sized to the same $f_c \approx 15.9$ Hz, so both configurations roll off at the same frequency and can be compared directly.

Because feedback reduces the diff pair's own input swing (V_{id} seen by M1/M2 is $V_{out,diff} / A_d$, not $V_{s,diff}$), the amplifier is kept operating over a much smaller, more linear portion of its transfer characteristic than it would in open loop — this is the practical reason closed-loop distortion is lower than open-loop distortion, and is worth confirming by comparing the input swing implied by A_d against the $\sim \pm 300$ mV linear range characterized in Experiment 4.

5. Safety Precautions

- Do not disturb the Experiment 4 bias network (R_D , R_{SS} , and the bias potentiometers) — verify the quiescent point (V_{cm} , V_{D1} , V_{D2} , $I_S \approx 80$ mA) is unchanged before adding the feedback network.
- Keep the function generator amplitude small enough that $V_{out,diff}$ stays within the linear range established in Experiment 4 (roughly ± 4 – 5 V around the 6.80 V

quiescent point) — start at a low amplitude and increase gradually while watching the oscilloscope for clipping.

- If using electrolytic (polarized) coupling capacitors, orient the positive terminal toward the higher DC side of each capacitor (toward V_{out} for the C_f branches, and check the function generator's DC offset for the C_{in} branches) — reversed polarity can fail short over time.
- Keep the oscilloscope probes' ground clips on a common circuit ground; do not ground two different non-ground nodes together through the probe grounds.
- Power off before changing any resistor or capacitor in the feedback network.

6. Procedure

6.1 Verify the Experiment 4 bias point

- With the feedback network disconnected, re-verify V_{cm} , $VD1$, $VD2$, and $I_S = V_{RSS}/R_S$ from Experiment 4. Confirm $I_S \approx 80$ mA before proceeding.

6.2 Unity-gain configuration

- Add $R_{in} = 10$ k Ω (with 1 μ F series C_{in}) from the function generator to each gate, and $R_f = 10$ k Ω (with 1 μ F series C_f) crossed as shown in Section 3. Add the 1 M Ω R_{bias} from each existing bias-pot wiper to the corresponding gate.
- Re-check $VD1$, $VD2$ with no input signal applied — they should be unchanged from Section 6.1, confirming the AC coupling has not disturbed the DC bias.
- Apply a small differential sine input (start at ~ 50 mV amplitude, 1 kHz) and increase gradually while watching V_{out} on the oscilloscope for clipping. Record the input and output amplitudes at a signal level well clear of clipping.
- Sweep the input frequency from about 5 Hz to 2 kHz, recording the output amplitude at each frequency, to trace the low-frequency roll-off.

6.3 Gain-of-10 configuration

- Power off. Replace R_{in} with 1 k Ω and C_{in} with 10 μ F on both input branches; leave R_f and C_f unchanged.
- Repeat the no-signal DC check, then the amplitude and frequency-sweep measurements as in Section 6.2, using a smaller input amplitude (start at ~ 10 – 20 mV) since the closed-loop gain is now higher.

7. Observation Tables

Table 1 — Closed-loop gain (1 kHz)

Configuration	$V_{in,diff}$ amplitude (mV)	$V_{out,diff}$ amplitude (V)	Measured gain $ A_v $
Unity gain	...	...	...
Gain of 10	...	...	...

Table 2 — Frequency response (repeat for each configuration)

Frequency (Hz)	$V_{out,diff}$ amplitude (V)	Gain $ A_v $	Gain (dB)
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8. Graphs to Be Plotted

- Gain (dB) versus frequency (log X axis), one curve per configuration — identify the -3 dB corner and compare with the predicted ≈ 15.9 Hz.
- Measured mid-band gain versus the ideal R_f/R_{in} and the loop-gain-corrected prediction (Section 9), for both configurations, on the same axes for direct comparison.

9. Theoretical Predictions

Using $A_d \approx 268$ V/V (Experiment 4's measured open-loop gain):

Configurat ion	R_f, R_{in}	$\beta = R_{in}/(R_f + R_{in})$	Loop gain $A_d \cdot \beta$	Ideal $A_v = -R_f/R_{in}$	Predicted actual A_v	f_c (Hz)
Unity gain	10 k Ω , 10 k Ω	0.500	134	-1.00	-0.993	15.9
Gain of 10	10 k Ω , 1 k Ω	0.091	24.4	-10.0	-9.61	15.9

10. Calculations — Matching Measured Results with Theory

- From Table 1, compute the measured gain magnitude for each configuration and compare with both the ideal R_f/R_{in} value and the loop-gain-corrected prediction in Section 9; compute the percentage deviation from each.
- From Table 2, identify the measured -3 dB frequency for each configuration and compare with the predicted 15.9 Hz.
- Back-calculate the loop gain implied by the measured deviation from ideal (using $A_{v,measured} = -A_d \cdot R_f / (R_f + R_{in}(1 + A_d))$, solve for A_d) and compare it with Experiment 4's measured A_d .

Quantity	Predicted	Measured	% deviation
Unity gain $ A_v $	0.993	...	...
Gain-of-10 $ A_v $	9.61	...	...
f_c , unity gain (Hz)	15.9	...	...
f_c , gain-of-10 (Hz)	15.9	...	...

11. Precautions

- Do not exceed the Experiment 4 device ratings ($V_{DS} = 20$ V, $V_{GS} = \pm 12$ V) at any point.
- Increase the input amplitude gradually in each configuration; back off immediately if the oscilloscope shows clipping at V_{out} .
- Power off before switching between the unity-gain and gain-of-10 component sets.
- Keep the function generator's DC offset at 0 V unless deliberately accounting for it in the R_{bias}/C_{in} divider.